

1 Ten Simple Rules for Implementing

2 Electronic Lab Notebooks (ELNs)

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10 S3 Appendix. Test questionnaire

11 This questionnaire is broadly based on the ZB MED ELN Guide (pp 45 to 49) [6].

12 Lab requirements

13 Does the ELN allow you to:

- 14 • Import and process special scientific formats?
- 15 • Export and process special scientific formats?
- 16 • Create your own templates (with drop-down menus to choose from a specified
17 vocabulary)?
- 18 • Import templates or offer preconfigured templates? If so, which formats are
19 supported?
- 20 • Share templates
- 21 ○ With members of your own research group?

22 ○ With external members who have a licence for the same ELN?

23 ○ With external members who have no ELN licence?

24 ● Link to files outside the ELN?

25 ● Work concurrently on the same documents as several other collaborators?

26 Does the ELN offer:

27 ● The special data entry and processing aids you need (e.g., scientific calculator,
28 formula editor, dictation devices, digital pen, animal module, antibodies module,
29 plasmids module, strains and cells module)?

30 ● Sample management and the possibility to create material databases within the
31 ELN?

32 ● Standard interfaces (e.g., connection to a Laboratory Information Management
33 System (LIMS) or other software)?

34 ● Collaboration functions (e.g., individual users and groups, role management)?

35 Usability

36 ● Is the ELN easy to navigate?

37 ● How easily can you access a folder you have uploaded and are now looking for?

38 ● Are the options self-explanatory?

39 ● Does the ELN allow you to insert images and tables?

40 ● Does the ELN allow you to use any text formatting you want?

41 ● Does the ELN allow for freehand drawing?

42 ● Does the ELN organise data in a way that meets your needs?

43 ● Are you confident the ELN will be easy to explain to senior lab members who are
44 used to physical lab notebooks?

45 ● Are you confident the ELN will be easy to explain to new lab members?

46 Good Research Practice (GRP)

47 Does the ELN allow you to:

- 48 • Search for keywords and experiments within the ELN using free text search,
49 advanced search, database queries and special search queries (e.g., searches for
50 chemical formulas and reactions)?
- 51 • Ensure evidentiary value (e.g., audit trail, electronic signature and time stamp)?
- 52 • Easily export backups? If yes, in which formats?
- 53 • Easily export the entire ELN (e.g., for archiving purposes, or to migrate it to another
54 ELN)?

55 Does the ELN offer:

- 56 • Documentation and traceability in line with GSP?
- 57 • Compliance with special regulatory requirements (e.g., GxP, ISO, FERPA, HIPAA)?
- 58 • Access (at least “read only”) for a minimum of ten years?
- 59 • (In-built) archiving / backups?

60 Connection to Research Data Management (RDM) system

61 Does the ELN allow you to:

- 62 • Assign persistent identifiers (e.g., Digital Object Identifiers (DOIs))?
- 63 • Reference an open experiment or sample (e.g., with a Persistent Identifier (PID)),
64 even if the folder structure changes in the future?

65 Does the ELN offer:

- 66 • Support for assigning metadata (e.g., input masks)?
- 67 • Automatic metadata harvesting (e.g., extraction from log files, export from Excel
68 spreadsheets)?
- 69 • Integration of controlled vocabularies?
- 70 • A connection to repositories, publishing platforms, file sharing services?
- 71 • A connection to digital preservation systems?